TaskVine / VineReduce / vine-cms-analysis-stack

Module 0 - Intro

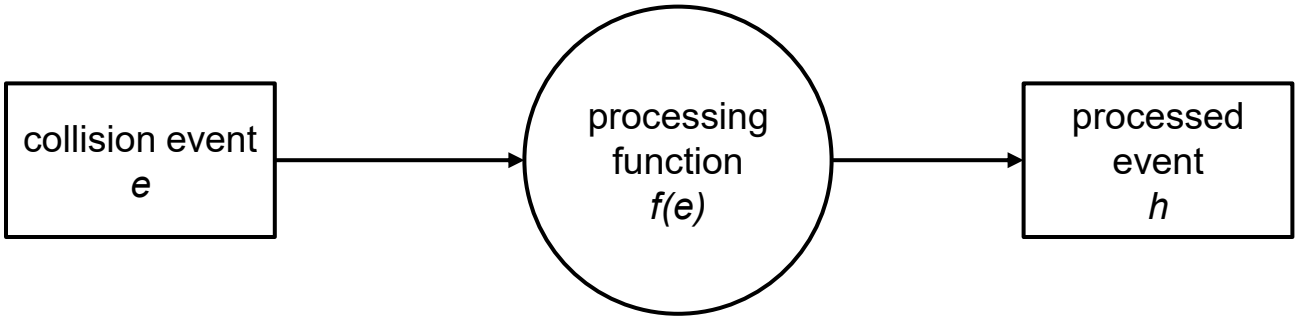
3-hour hands-on tutorial for HEP graduate students

Scaling from First Principles

Apply processing function **f** to event **e** to get result **h**.

f: E → H

h = f(e)

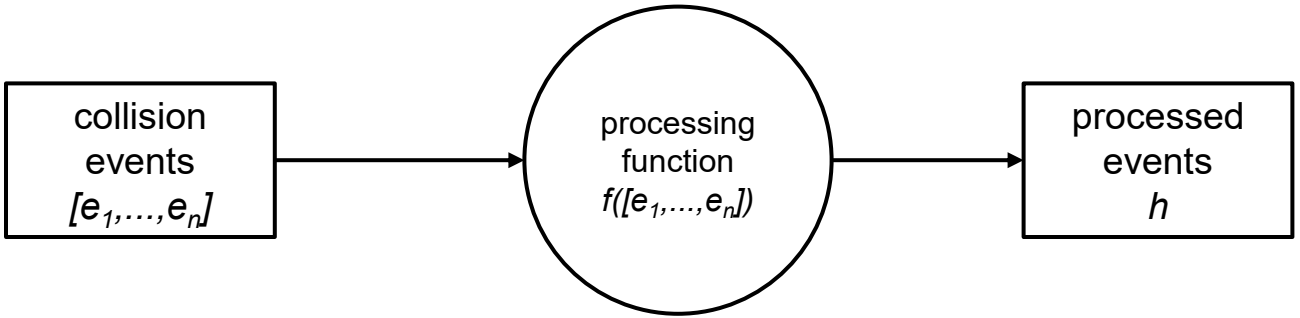


Scaling from First Principles

Almost always we process more than one event per function call.

$f: E \rightarrow H, \quad +: H \times H \rightarrow H$

$f([e_1,...,e_n]) = f(e_1) + ... + f(e_n)$



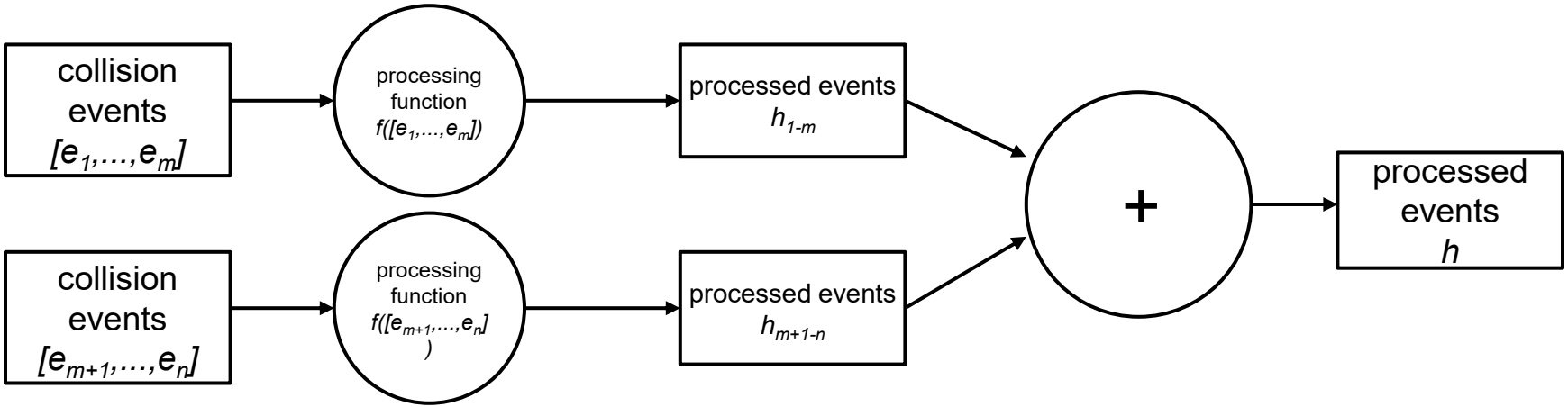
If $f([e_1,...,e_n])$ can run efficiently in a single machine, we are done.

Scaling from First Principles

The + only comes into play when we want to scale the computation, and only if it is associative.

(It is like map-reduce, but the reduce part is only important when it allows us to scale.)

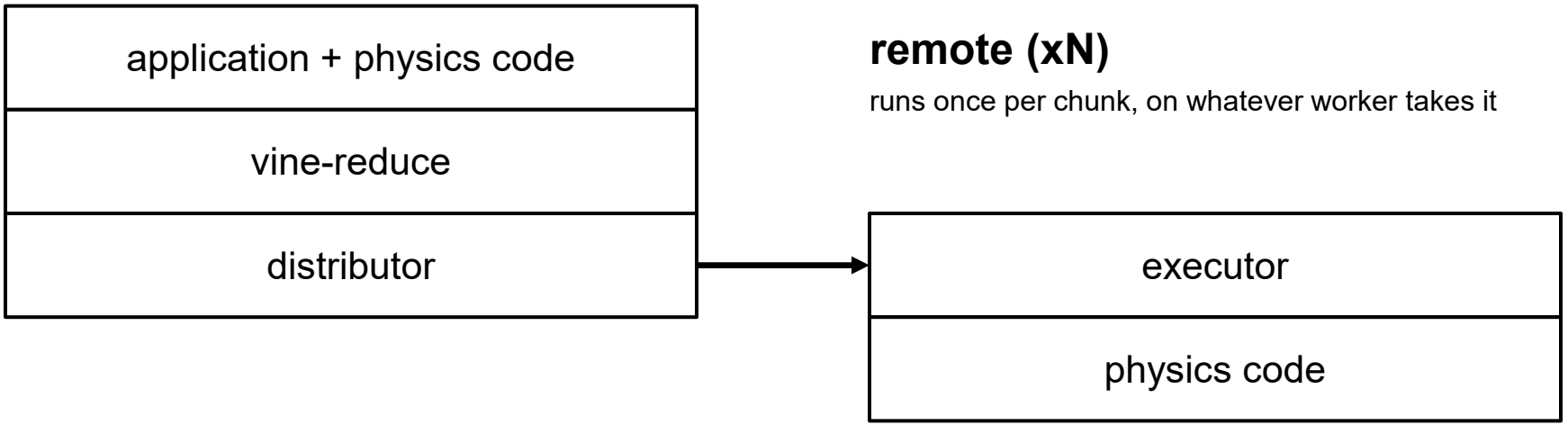
$f([e_1,...,e_n]) = f([e_1,...,e_m]) + f([e_{m+1},...,e_n]) = f(e_1) + ... + f(e_n)$



The shape of the stack

local (x1)

runs once, on the machine you are logged into



What we are running

- ADL benchmark Q6: a top-quark trijet selection on CMS Open Data
- Small, but the same shape as a real
- Written in coffea or RDataFrame; you pick one



Partial results are added together until one result per dataset remains.

Why VineReduce, alongside coffea/RDataFrame

Parallelism

Owned by VineReduce

- Split a dataset into chunks
- Map the processor over chunks
- Reduce outputs back together

Distribution

Handed to a distributor (today: TaskVine)

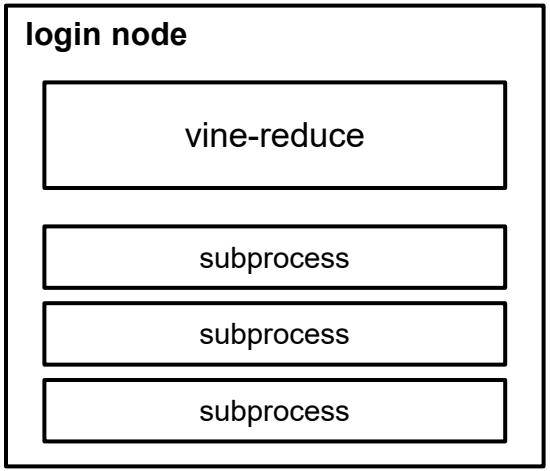
- Schedule chunks onto machines
- Transfer code and data
- Run on remote workers

The analysis code never changes. Only what it is handed to does.

Three places for execution

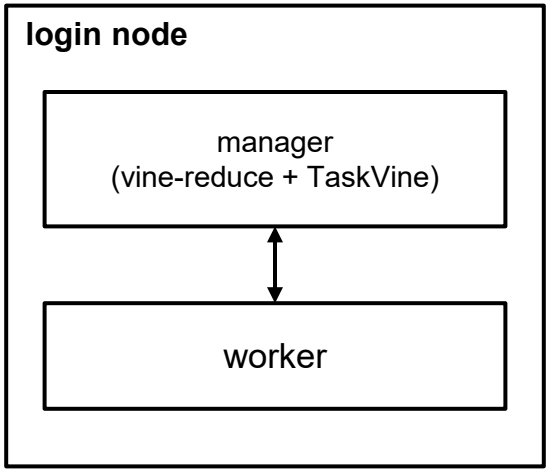
1 Local distributor

Module 2



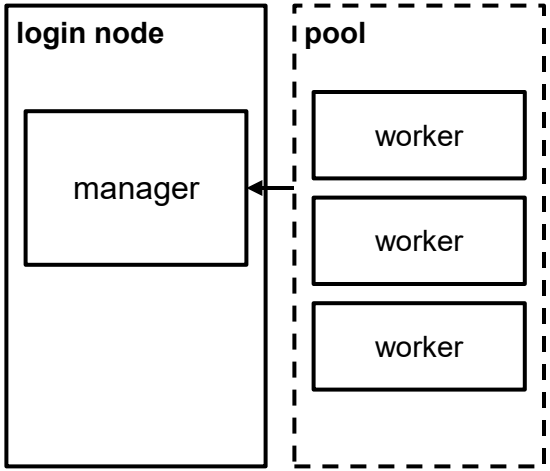
2 Local TaskVine

Module 3



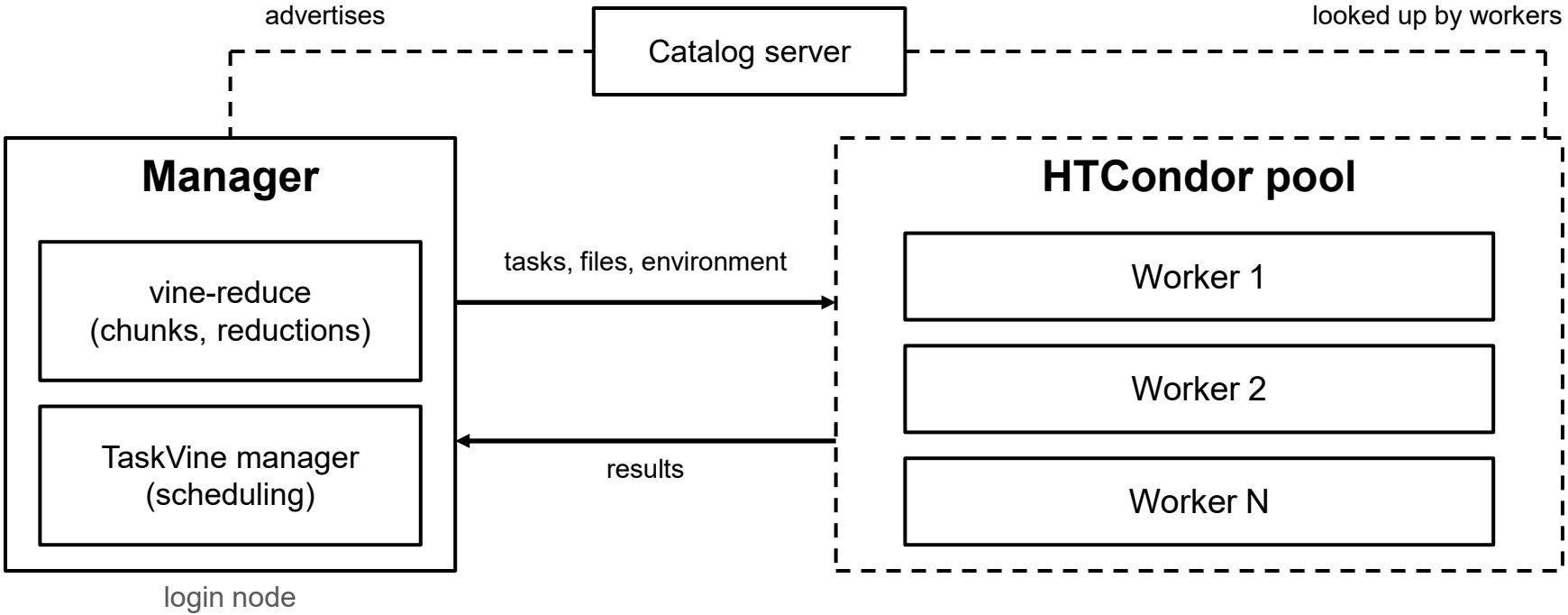
3 HTCondor pool

Module 4



The analysis code is the same in all three. Only what it is handed to changes.

Manager and workers



Workers find the manager through the catalog, connect to it, and read input data from storage (xrootd or a shared filesystem).

What the manager lets you control

Module 5	**Resources and chunk size**	What a task asks for, and how big it is
Module 6	**Checkpoints and restart**	Stop a run, resume without redoing work
Module 7	**result_postprocess**	Big outputs bypass the manager
Module 8	**Multiple processors**	Several analyses in one run
Module 9	**failure_proportion**	Keep going when some chunks fail

Where to get help

NDCMS Mattermost

`https://mattermost.web.cern.ch/ndcms/channels/cctools`

Cooperative Computing Lab

`https://cclnd.cse.nd.edu`

TaskVine documentation

`https://cctools.readthedocs.io/en/stable/taskvine`

GitHub

`github.com/cooperative-computing-lab: vine-reduce, vine-cms-analysis-stack, cctools`